GAME DESIGN & LEVEL SPECIFICATION DOCUMENT (GDD)

An Echo Has No Shape

Single-Verb First-Person Exploration & Acoustic Level Design Specification

Author / Role	Kevin (Lefi) Shan: Solo Game & Level Designer, Programmer	**Engine & Stack**	Unity 6 (URP), C#, Custom HLSL Spherical Wave Shaders
Production Scope	4-Week Solo Prototype · USC CTIN (Situated Role: The Traveler)	**Core Constraints**	1 Primary Action (Press E) · Zero HUD / Minimaps · ~550m Single Scene

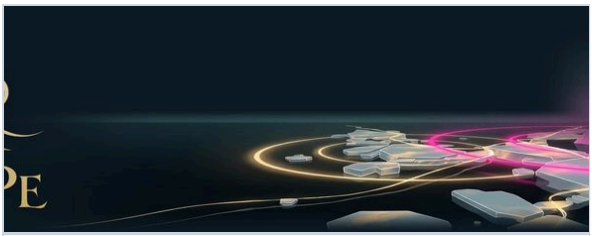


Figure 1.1: Player's white echo pulse colliding with the Siren's magenta wave in the Glacial Ocean.

Executive Summary & Level Design Goal

Many echolocation games treat sound as a simple flashlight in the dark: once the pulse reveals the room geometry, the core mechanic stops evolving because the relationship between player input and returned information stays the same.

To focus a 4-week solo cycle on level design and systemic puzzle variety, An Echo Has No Shape locks the player to a single primary action (Press E to Emit Echo, plus F to Interact) across a continuous ~550m x 550m fog-covered basin. Progression is driven not by unlocking new abilities, but by changing how each region responds to the player's pulse.

1. Core Level Design Pillars & Single-Verb Framework

Section 1.1 to 1.3

1.1 Exploration Through Pacing (Traveler)

Rather than piecing together text logs, the player reads the world through movement, positioning, and timing. Where you stand when you emit a pulse, and when you stop to listen, directly control which paths open.

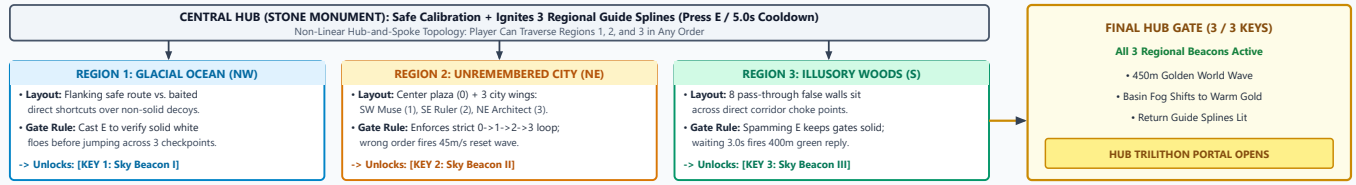
1.2 Single-Verb Constraint (Press E)

The player's move set never expands: Emit Echo (E, 42m range, 5.0s cooldown) and Interact (F at stone tablets). Removing combat, inventory, and upgrades forces every regional puzzle to come from level layout and signal behavior.

1.3 Three Regional Rule Variations

Across the three spokes, the level design changes what the player's echo accomplishes: verifying solid platforms against decoy signals (Region 1), unlocking a 3-wing city in sequence (Region 2), and pausing 3.0s so a forest reply opens choke-point gates (Region 3).

Figure 1.2: Single-Verb Recontextualization & Regional Gating Progression (Hub -> Regions 1/2/3 -> Finale) Same Player Action (Press E) -> Three Distinct Spatial Gating Rules -> 3/3 Beacons Unlock Final Gate



2. Regional Level & Puzzle Rule Matrix

Table 2.1: Regional Rulesets & Gating

Design Axis	Hub: Stone Monument	Region 1: Glacial Ocean	Region 2: Unremembered City	Region 3: Illusory Woods
Level Topology	**Central Hub Ring** 3 spoke heads + final gate	**Branching Floe Lanes** Flanking arc vs. direct shortcut	**3-Wing Street Grid** Center plaza + SW / SE / NE wings	**Corridor Maze** Choke gates vs. detour loops
Habit Tested	Navigating without permanent sight or UI markers.	Jumping onto every lit platform along a shortcut.	Activating landmarks in random order or leaving early.	Spamming E immediately whenever the 5.0s cooldown ends.
Environmental Signal	**Prophecy Tablet** Ignites 3 guide splines	**3 Siren Emitters** 62m pink wave every 7.0s	**3 Wing Statues** 150m blue / red / green waves	**Forest Reply Center** 400m green wave after 3.0s wait
Fail State / Reset	Safe zone (zero hazards or reset states).	Stepping on a pink decoy floe respawns at last tablet.	Wrong order or exiting bounds fires 45m/s reset wave.	Pressing E early overwrites green wave & shuts gates.
Tablet Clue	*"Three silences wait in the mist..."*	*"Not every voice across the ice wishes you to reach the shore."*	*"The city stands whole within the meeting of many eyes."*	*"The woods have opened to the voice you left room to hear."*

3. Hub-and-Spoke World Layout & 4-Layer Sightless Wayfinding System

`~550m x 550m Single-Scene Basin (Main.unity)`

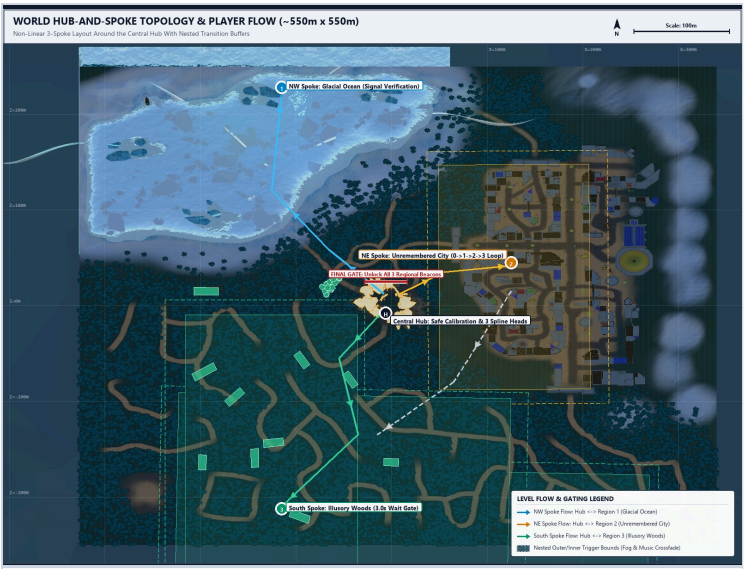


Figure 3.1: World Hub-and-Spoke Flow & Wayfinding Map (~550m x 550m): Showing the Central Hub (H), the 3 spoke traversal flows (1, 2, 3), secondary inter-regional loops, and nested fog transition bounds.

3.1 Non-Linear Hub-and-Spoke Routing

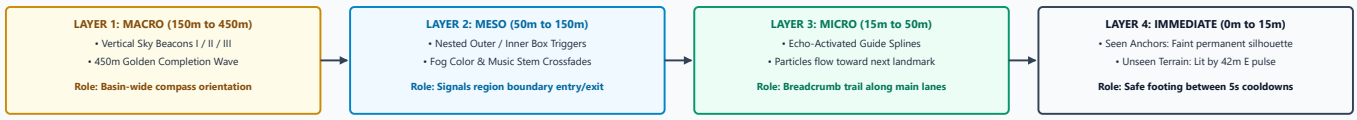
The entire ~550m x 550m basin is built inside a single continuous scene around the Central Hub. After interacting with the hub tablet to light the three guide spline heads, players can tackle the three spokes in any order:

- **NW Spoke (Glacial Ocean):** Leads from the western shore across three checkpoint islands to the northern beacon.
- **NE Spoke (Unremembered City):** Crosses the west bridge into a 200m x 220m walled street grid with an outer reset boundary.
- **South Spoke (Illusory Woods):** Has two entry points: the NW hub path (Tablet T1) and a secondary SE path from the city outskirts (Tablet T2).

3.2 Nested Trigger Volumes for Smooth Transitions

Because heavy fog hides the distant horizon, each region is surrounded by nested OuterBounds and InnerBounds box triggers. Walking through the 40m buffer zone between the outer and inner bounds smoothly crossfades the fog color, fog view distance, and regional music track so players immediately feel when they enter or leave a region.

Figure 3.2: 4-Layer Wayfinding Hierarchy & Hub-and-Spoke Return Loop (Zero HUD / Zero Minimap) How 4 Distance Bands Guide the Player From Macro Landmarks Down to Immediate Footing



3.3 Four-Layer Wayfinding Specification Table

`Table 3.1: Spatial Orientation Layers`

Wayfinding Layer	Range	Level Design Mechanism	Technical Implementation	Player Navigation Purpose
1. Macro Scale	`150m to 450m`	**Sky Beacons & World Wave**	Completing a region fires a 450m gold wave (12s hold) and activates its sky beacon.	Gives players a high-altitude landmark visible above the fog from anywhere in the basin.
2. Meso Scale	`50m to 150m`	**Nested Fog & Audio Bounds**	Blends fog color, view distance, and music tracks across the 40m Outer/Inner buffer.	Tells the player they are entering or leaving a region before geometry appears.
3. Micro Scale	`15m to 50m`	**Breadcrumb Guide Splines**	Spline ribbons stay invisible until hit by an echo, then stream particles forward.	Guides players between major checkpoints without cluttering the ground with permanent arrows.
4. Immediate Scale	`0m to 15m`	**Anchor Silhouettes vs. Echo Terrain**	Splits materials into Seen (faint silhouette in fog) and Unseen (0% visibility until echoed).	Keeps key checkpoints (tablets, statues, spires) visible even while E is on cooldown.



Figure 3.3: Hub Initial State (0/3 Beacons): Player echo illuminating the monolith ring and central tablet.

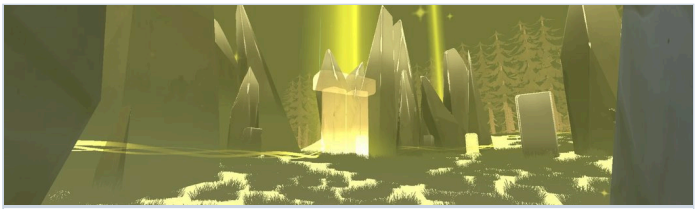


Figure 3.4: Hub Finale State (3/3 Beacons): All 3 regional beacons active and the Golden Gate unlocked.

4.1 Region 1 (Glacial Ocean): Flanking Safe Routes vs. Baited Decoy Shortcuts

`Puzzle Rule: Signal Verification Before Platforming`

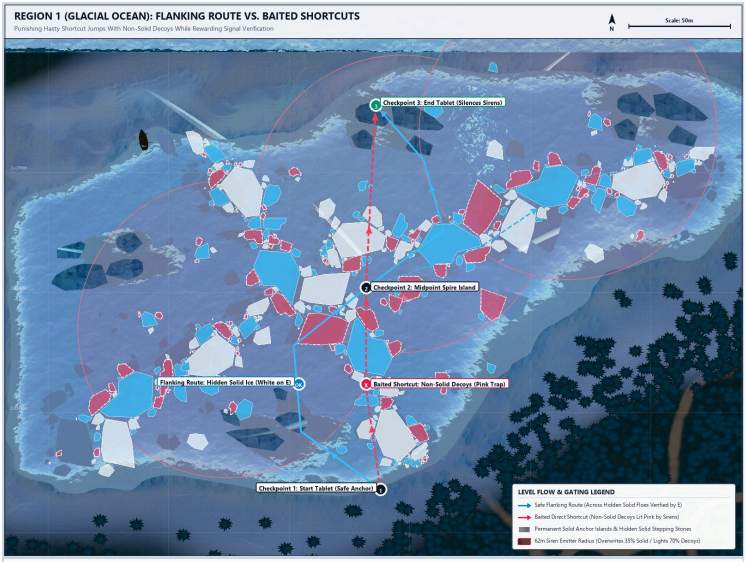


Figure 4.1: Region 1 Flow & Platform Classification Map (Glacial Ocean): Showing the Safe Flanking Route (blue arrows) across solid/hidden-solid ice vs. the Baited Direct Shortcuts (red dashed arrows) over non-solid Siren decoys.

Level Design Goal: Decision Tension Over Jump Precision

First-person platforming over water becomes repetitive if every visible platform is safe, and frustrating if jumps require pixel-perfect timing. The Glacial Ocean builds tension through route verification. Three Siren emitters (A, B, C) fire a 62m magenta pulse every 7.0s that corrupts nearby ice colors and highlights non-solid decoy floes along tempting direct shortcuts.

Three Platform Types & Signal Responses

- **1. Permanent Anchor Islands (White on Map):** Large islands and checkpoints (1 Start Tablet, 2 Midpoint Spire Island, 3 End Tablet) that stay faintly visible in the fog as safe vantage points.
- **2. Hidden Solid Stepping Stones (Cyan on Map):** Solid platforms with 0% base visibility arranged along the flanking route. The player's E pulse always reveals them in pure white (#FFFFFF); a Siren wave hides 40% in fog and turns 35% pink.
- **3. Non-Solid Decoy Traps (Pink Dashed on Map):** Water hazards with no collider placed along direct shortcuts. Player echoes pass straight through them (0% return), but Siren waves light 70% of decoys in bright pink (#FF0F6B).

Baited Shortcut vs. Flanking Route Layout

Between Checkpoints 1 -> 2 -> 3, the shortest straight line is lined with non-solid decoys inside the 62m Siren radii. Because a pink platform could be either a 35% overwritten solid floe or a 70% decoy trap, players learn to pause on safe islands and press E to verify white solid geometry along the flanking route.

Figure 4.2: Region 1 Lane Routing & Platform Signal Verification Matrix Why Players Must Cross-Check Pink Platforms With Their Own White Echo (Press E) Before Jumping



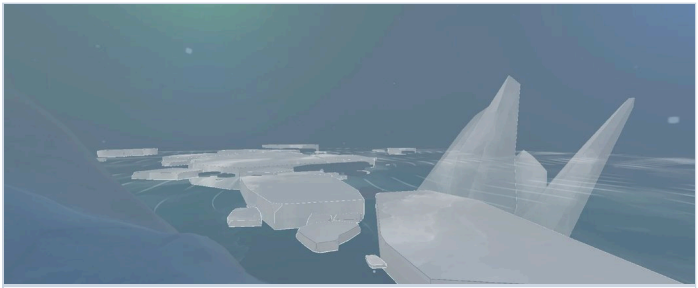


Figure 4.3: White Echo Verification: Player echo revealing solid white stepping stones before the twin spires.



Figure 4.4: Region 1 Completion State: Interacting with Checkpoint 3 silences the Sirens and lights the route in gold.

4.2 Region 2 (Unremembered City): 3-Wing Sequential Loop & Reset Boundary Gate

`Puzzle Rule: 0 -> 1 -> 2 -> 3 Cumulative Architectural Layering`

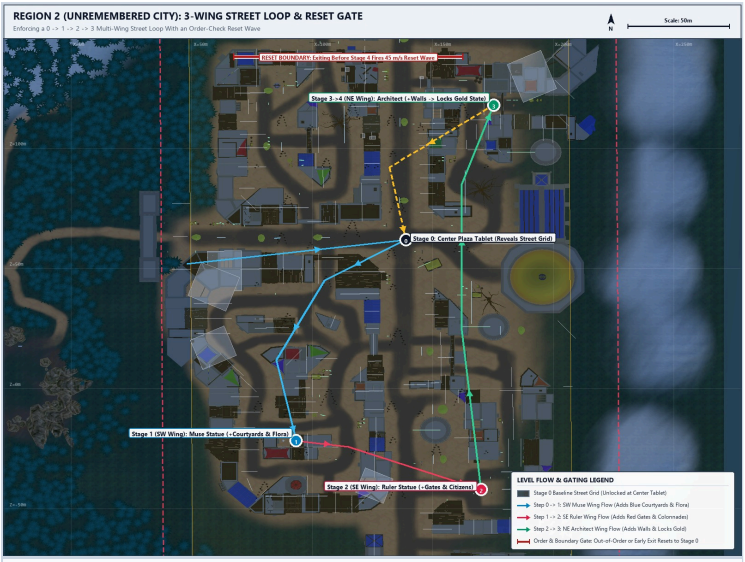


Figure 4.5: Region 2 Street Loop & Reset Boundary Map (Unremembered City): Showing the required street traversal loop from Center Plaza (0) -> SW Muse Wing (1) -> SE Ruler Wing (2) -> NE Architect Wing (3) and the outer Reset Boundary.

Level Design Goal: Multi-Wing Spatial Loop & State Layering

In the Unremembered City, each landmark reveals only one architectural layer of the ruined street grid. No single statue restores the city on its own. To unlock the region, the player must navigate a full clockwise loop across all three city wings in strict sequence (0 -> 1 -> 2 -> 3), layering each wing's geometry onto the city until the structure locks into its completed state.

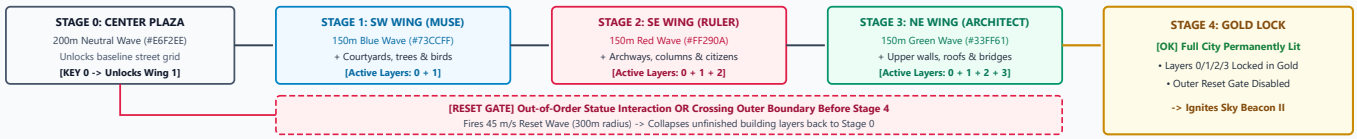
Cumulative Architectural Layers (Stages 0 to 4)

- **Stage 0: Center Plaza Tablet (Neutral Wave #E6F2EE):** Fires a 200m pulse that reveals the baseline street grid and low foundation walls across the plaza.
- **Stage 1: SW Wing Statue, The Muse (Blue Wave #73CCFF):** Shifts the fog to sky-blue and adds courtyard gardens, trees, and bird silhouettes onto the Stage 0 streets.
- **Stage 2: SE Wing Statue, The Ruler (Red Wave #FF290A):** Shifts the fog to crimson and adds grand archways, colonnades, and citizen figures on top of Stages 0 and 1.
- **Stage 3 -> 4: NE Wing Statue, The Architect (Green #33FF61 -> Gold #FFCC4D):** Adds upper walls, roofs, and bridges, permanently locking all layers into Stage 4 Gold Completion.

Sequence Gate & 45 m/s Reset Boundary Wave

Interacting with a wing statue out of order (anything other than 0 -> 1 -> 2 -> 3) or walking past the outer city boundary before reaching Stage 4 triggers a 45 m/s Reset Wave from the center plaza that wipes unfinished building layers back to Stage 0.

Figure 4.6: Region 2 Lock-and-Key Sequence Chart (0 -> 1 -> 2 -> 3 -> Gold Lock) & Reset Gate Each Wing Statue [KEY] Requires the Prior Stage Active and Preserves All Earlier Building Layers



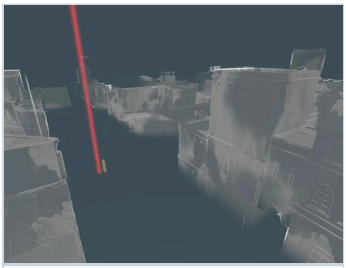


Figure 4.7: Stage 0 (Center): Baseline street grid and center tablet.



Figure 4.8: Stage 1 (SW Muse): Sky-blue fog, courtyards, and trees.



Figure 4.9: Stage 2 (SE Ruler): Crimson fog, archways, and citizens.

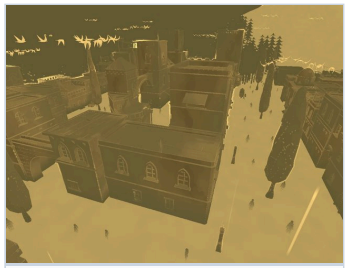


Figure 4.10: Stage 4 (Complete): All building layers locked in gold.

4.3 Region 3 (Illusory Woods): Corridor Choke-Point Gates & 3.0s Silence Window

`Puzzle Rule: Input Restraint & 3.0s Delayed Forest Reply`

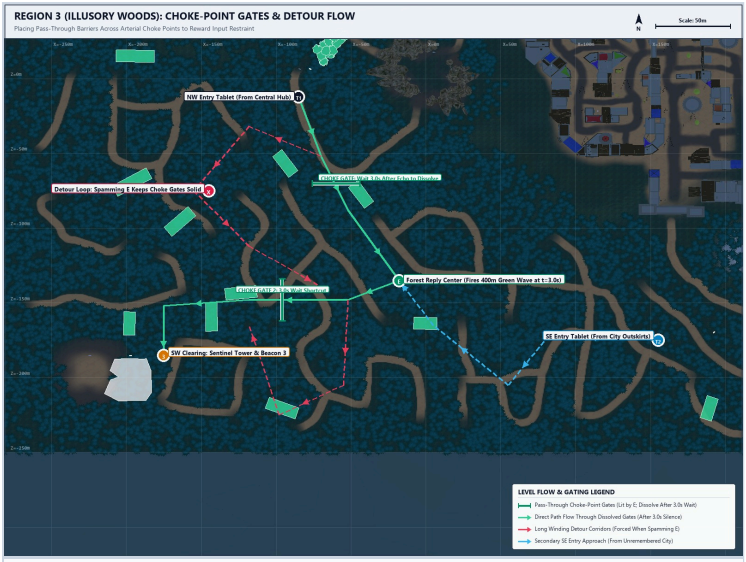


Figure 4.11: Region 3 Choke-Point Gates & Detour Flow Map (Illusory Woods): Showing the Direct Path (green arrows) through pass-through choke-point gates after a 3.0s wait vs. the Winding Detour Loops (red dashed arrows) when spamming E.

Choke-Point Gate Rule: Player Echoes Keep the Barrier Solid

Across the main junctions leading to the SW Sentinel Tower, 8 pass-through illusory walls sit across the dirt paths. When the player casts a white echo (42m range, 1.4s duration), the pulse lights up these false walls with the exact same white shader as solid corridor trees, making the direct path look like a dead end and funneling impatient players into long detour loops.

Unlocking the Choke Gates: If the player casts an echo and then waits 3.0s without pressing E again, the center of the forest answers with a 400m Green Reply Wave (50 m/s) that dissolves all false walls for 5.0s so the player can walk straight through. Pressing E during this open window triggers a 0.35s wave overwrite that cuts off the green wave and snaps the gates shut.

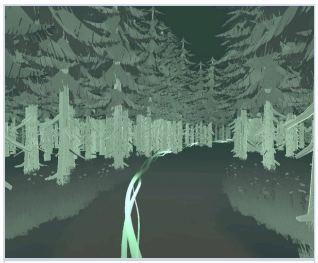


Figure 4.12: Corridor Spline: Dirt corridor lit by an echo-activated spline.

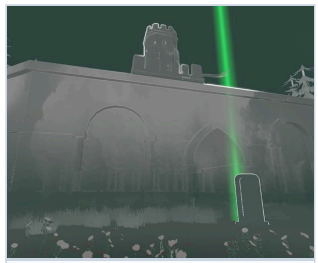
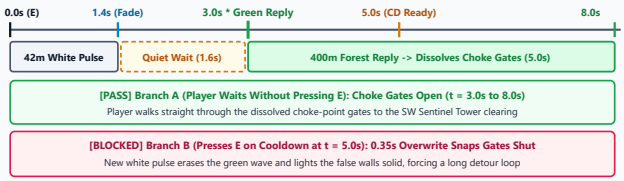


Figure 4.13: SW Sentinel Tower: Final clearing reached after passing the gates.

Table 4.1: Playtest Tuning of the 3.0s Wait Window (6 Cohorts)

Wait Time	Accidental	Solved Rate	Observed Playtest Behavior
`1.0s`	85%	15%	Players trigger green reply while walking without noticing why.
`2.0s`	45%	50%	Mixed results: fast walkers still trigger it by accident.
`3.0s` *	**10%**	**90%**	**[Tuned Sweet Spot] White pulse fades at 1.4s; leaves 1.6s quiet gap before 5.0s cooldown.**
`4.0s`	5%	60%	Impatient players start pressing E again just before cooldown ends.
`5.0s`	0%	20%	Collides with 5.0s cooldown: players press E on habit, resetting timer.
`6.5s`	0%	5%	Too long: players assume nothing happened and keep echoing.

Figure 4.14: 8-Second Input Timing Timeline & Choke-Gate State Switch Wait 3.0s -> Gate Opens vs. Spam E -> Gate Snaps Shut



5. Level Design Retrospective & Key Takeaways

`Post-Mortem Synthesis`

5.1 Recontextualizing One Verb by Region

Keeping the player on a single E input meant puzzle variety had to come from level design: platform verification in Region 1, sequential 3-wing routing in Region 2, and timed choke-point gates in Region 3.

5.2 Visual State Feedback Over Text Hints

In Region 3, tablet hints alone did not stop players from spamming E on cooldown. Adding the 0.35s wave overwrite (where pressing E visibly cuts off the green wave and snaps the gate shut in front of the player) raised solve rates from 15% to 90%.

5.3 Matching Floorplans to Puzzle Rules

Each region's layout reinforces its mechanic: non-solid decoys placed along tempting direct shortcuts in the ocean, three statues separated across city wings, and pass-through walls placed across main forest choke points.